

SHUVAM PAL

+91 85830 54679 ishuvam.pal@gmail.com Shuvam Pal unichronic

Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built a resilient automation backend around public-site workflows with retries, session recovery, API lookups, and state validation, reducing manual filing time by **85%**.
- Used browser automation to exercise dynamic portal flows, capture failure states, and validate form/session behavior before handing work back to operators.
- Containerized services with Docker, deployed to Google Cloud Run, and automated releases through GitHub Actions, improving deployment velocity by **70%**.

Google Summer of Code

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built Python medical-image segmentation tooling for **95 brain subparts**, handling preprocessing, orientation correction, model inference, generated masks, and large-volume MRI data paths.
- Reduced segmentation runtime by **87%** using asynchronous and parallel execution while preserving inspection workflows for generated masks and 3D surfaces.
- Debugged pipeline regressions across conformation, orientation, label mapping, PyTorch/ONNX-style inference, and binary mask generation in a mature open-source codebase.

Open Source

[Contributions](#)

- Contributed patches across Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working inside maintainer review, CI, reliability, and integration constraints.

Projects

Murdoc | Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker

[GitHub](#)

- Built an AI gateway and **behavior harness** for LLM, HTTP-tool, and MCP traffic with route profiles, request scoring, runtime policy checks, and allow/block decisions before downstream execution.
- Added configurable test-lab APIs, attack-corpus runs, audit exports, OpenTelemetry traces, Prometheus metrics, and health checks so regressions and unsafe tool behavior were visible early.

Postificus | Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus

[GitHub](#)

- Built REST API and worker services for content publishing with direct API paths, Go-Rod browser automation fallbacks, shared browser sessions, and recoverable job state.
- Added API/controller tests, retry/DLQ handling, circuit-breaker isolation, health checks, and Prometheus metrics for unreliable external API and browser workflows.

Swish Support System | Python, Go, FastAPI, Gin, PostgreSQL, Redis, Langfuse, Pytest

[GitHub](#)

- Built an AI-assisted support backend where LLM calls classify complaint intent, severity, ambiguity, and evidence usefulness while deterministic services control order/refund/escalation policy.
- Built regression tests and LLM-judge checks for response quality, copy-contract preservation, policy safety, session context, evidence handling, and multi-turn support flows.

Technical Skills

Languages: Python, Go, TypeScript/JavaScript, C++

Frameworks: FastAPI, Gin, Echo, Node.js, Pytest, Go test

Tools: REST APIs, Docker, GitHub Actions, PostgreSQL, Redis, RabbitMQ, Go-Rod, Puppeteer, MCP, Prometheus, OpenTelemetry, Langfuse, Codex, Claude, Cursor